

CNN Vulnerability On Untargeted Adversarial Attacks

1st Aleksandar Stankovic

Singidunum University

Belgrade, Serbia

aleksandar.stankovic.23@singimail.rs

2nd Marina Marjanovic

Singidunum University

Belgrade, Serbia

mmarjanovic@singidunum.ac.rs

Abstract—Convolutional Neural Networks (CNNs) are widely used in various AI-driven industries, including healthcare, autonomous systems, and cybersecurity, due to their high accuracy in image classification. However, these models are vulnerable to adversarial attacks, where small, imperceptible perturbations in input data can cause incorrect predictions. This study evaluates the impact of five adversarial attacks: Fast Gradient Sign Method (FGSM), Iterative FGSM (IFGSM), Momentum Iterative FGSM (MIFGSM), Deepfool, and OnePixel Attack on the accuracy of a CNN trained on the CIFAR-10 dataset. Each attack is analyzed for its unique influence on model performance, revealing significant accuracy drops under adversarial conditions. The results are analyzed and illustrate the potential security risks of adversarial attacks, underscoring the need for robust defenses in CNN based systems.

Index Terms—neural network, adversarial attacks, model accuracy

I. INTRODUCTION

Artificial Intelligence (AI) technologies, particularly Convolutional Neural Networks (CNNs), are increasingly critical in industries such as healthcare, finance, and transportation. Machine Learning (ML), a central AI methodology, enables high-accuracy image classification and pattern recognition across diverse datasets. However, CNNs are vulnerable to adversarial attacks—small, targeted changes to input images that lead to misclassifications. In high-stakes applications like self-driving cars or medical diagnostics, these vulnerabilities can have serious consequences. This study explores the impact of various adversarial attack techniques on CNN model accuracy, aiming to provide insights into attack types and highlight the importance of robust defenses in ML systems. Pioneers in this field, Szegedy et al. in 2014, [1] and I. J. Goodfellow in 2015, [2] show that back then state-of-the-art machine learning models were vulnerable to adversarial examples. Their point of view was that, adding an imperceptibly small vector, which is equal to the sign of the gradient of cost function with respect to the input, we could change model prediction accuracy and often significantly altered image classification. In this research we propose several types of adversarial attacks on convolutional neural network models. The attacks are FGSM [2], MIFGSM [4], IFGSM [5], DeepFool [6] and Onepixel

attack [7]. Mainly, these attack algorithms are designed and operate in the white-box environment [8]. Mentioned attacking algorithms are performed on simple CNN architecture for CIFAR10 dataset, and based on adversarial goal using untargeted adversarial attack method [9]. In untargeted attacks, the aim is to enforce the model to misclassify the adversarial image, regardless of the output. Considering that all methods based on adversarial goals will be widespread in the future, the aim was to present the results and show how different types of untargeted adversarial attacks affect the accuracy of the model.

II. THE DATASET AND CONVOLUTIONAL NEURAL NETWORK (CNN)

A. The dataset – CIFAR10

The CIFAR-10 dataset consists of 60000 (32x32 resolution) colour images sorted into 10 classes, with 6000 images per class. Dataset is split on 50000 training images and 10000 test images. All these images are in the form of pixels and represent 32 x 32 pixel values for 3 different channels (RGB). Based on that, as input data for CNN we are going to use 60,000 images of (32,32,3) shape and 10 output classes to be classified into.

B. Convolutional neural network (CNN)

Taking for i.e. one image from CIFAR10 dataset with resolution 32x32 pixels we get number of 1024 neurons per image and considering that with RGB comes depth of 3 channels (colors) we have 3072 single neurons by image in input layer. The idea of using CNN for image classification lies in that they are comprised of neurons that self-optimize through learning [10][11]. The CNN model consists of a sequence of layers designed to extract features from the input images. The architecture includes an initial 2D convolutional layer with 32 filters of size 3x3, followed by a ReLU (Rectified Linear Unit) activation function to introduce nonlinearity. A max-pooling layer with a 2x2 window reduces spatial dimensions by half. After several convolutional and pooling layers, the network includes dense (fully connected) layers and concludes with a Softmax layer, which outputs the probability distribution over the 10 classes in CIFAR-10 [12, 13].

During the training process of a machine learning model, we could indicate the model's performance over 10 training epochs (cycles).

This research was supported by the Science Fund of the Republic of Serbia, Grant No. 7502, Intelligent Multi-Agent Control and Optimization applied to Green Buildings and Environmental Monitoring Drone Swarms - ECOSwarm.

Fig. 1 plots the loss function value of the model on the training set and the validation set over epochs, and measures how well the model is performing. Training and validation loss decrease as the number of epochs increases, which suggests that the model is learning and improving its predictions over time. Validation loss slightly increases towards the end, which could be an indication of the model beginning to overfit to the training data, so we decided to stop on 10 steps because simplicity of CNN.

Fig. 2 shows the accuracy of the model both on the training and validation set over epochs. Graph shows that the accuracy increases on both the training and validation sets as the number of epochs increases.

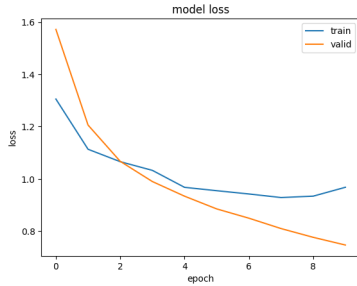


Fig. 1. Model loss.

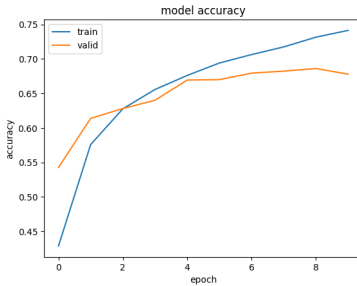


Fig. 2. Model accuracy.

III. ATTACK METHODS

We study a variety of adversarial attack methods including characteristics as well as the capabilities and goals of the adversarial attacks:

A. Fast Gradient Sign Method (FGSM)

FGSM is a basic, widely used method that aims to cause misclassification by adding a calculated perturbation to the input image. Its simplicity makes it a foundational approach in adversarial attacks, often serving as a baseline to evaluate model robustness. FGSM has misclassification as its main goal and includes several steps: calculate the loss function, calculate the gradient with respect to the pixels of the image and in the direction of the gradients maximize the loss function calculated above. In our image classification task, this perturbation is designed to be visually imperceptible but can

cause high-confidence misclassifications. Adversarial examples using FGSM are produced by the following mathematical formula by Eq. (1):

$$X^{adv} = X + \varepsilon \cdot \text{sign}(\nabla_x J(X, y_{true})) \quad (1)$$

Regarding adversarial example in this equation, X is the original image, y_{true} is the true target class associated with original image, ε denotes the parameter which controls the amplitude of perturbation. J is the loss function of the current CNN structure and ∇_x is the set of parameters in this model [5, 14].

B. Momentum Fast Gradient Sign Method (MFGSM)

The momentum method is an attacking method that enhances gradient by introducing a speed vector and aligns with the gradient direction of the loss function between iterations. MIFGSM introduces a momentum term to the iterative process, making it less likely to be trapped in local gradients and increasing its effectiveness in real-world applications. MIFGSM's momentum-based adjustment makes it particularly effective against models with gradient-smoothing defenses. It generally outperforms FGSM and IFGSM in terms of misclassification rates and the stealth of perturbations. This addition aims to stabilize the update directions and prevent getting stuck in suboptimal local maxima. Adversarial examples using FGSM are produced by the following mathematical formula by Eq. (2):

$$X_{n+1}^{adv} = \text{Clip}_x^\epsilon \{x_n^{adv} + \mu \cdot \text{sign}(g_n + 1)\} \quad (2)$$

where decay factor μ controlling the momentum term and the accumulated gradient g guiding the direction of updates at each iteration n [4].

C. Iterative Fast Gradient Sign Method (IFGSM)

IFGSM builds on FGSM by applying it iteratively, allowing the model to “refine” the perturbation with each step to maximize the model’s error further. This often results in a stronger, more effective attack. The IFGSM applies FGSM multiple times iteratively with a small step size α and can be expressed by the following mathematical formula by Eq (3):

$$x_0^* = x, x_{t+1}^* = x_t + \alpha \cdot \text{sign}(\nabla_x J(x_t, y)) \quad (3)$$

By applying multiple small steps, IFGSM makes it easier to evade detection while maintaining a significant impact on model accuracy. The difference from FGSM is that here in each iteration adjusts the perturbation to maximize the loss further [5, 15].

D. Deepfool attack

Deepfool was designed to iteratively compute the smallest possible adversarial perturbation for a specific image, aiming to locate the nearest decision boundary to the original sample x and identify the minimal adversarial examples just across that boundary. This method addressed the challenge of selecting the parameter ε in the FGSM method and executed the attack on general non-linear decision functions using multiple linear

approximations [6]. Deepfool is highly efficient in that it creates minimal changes to the image, often resulting in perturbations that are imperceptible, making it challenging to detect visually. Calculating minimum perturbation can be expressed by the following mathematical formula by Eq (4):

$$-\frac{f(x_0)}{\|\omega\|_2^2} \cdot \omega \quad (4)$$

E. Onepixel attack

The OnePixel attack alters only a single pixel within the image, challenging the robustness of a model with minimal, localized changes. The sub-image is then compared with the parent image to retain the sub-image with the best attack effect according to the selection criteria. This process is repeated until the desired level of misclassification is achieved. This approach demonstrates that even tiny perturbations altering only one pixel can impact model predictions. This finding is particularly important for applications that rely on high-resolution images, as it shows that models may be sensitive to even slight disruptions. Generate adversarial examples by only perturbing the input with a minimum of one pixel can be expressed by the following mathematical formula by Eq (5):

$$\text{maximize } f_{adv}(x + e(x)) \leq d \quad (5)$$

In above formula x refers to the original input vector, $e(x)$ refers to the perturbation, d is the number of perturbed pixels, and f_{adv} is the confidence of the target class.

IV. EXPERIMENTAL RESULTS

Based on systematic literature review, we selected five untargeted adversarial attacking techniques to evaluate their influence on the accuracy of the CNN model. The adversarial attacks are FGSM (Fast Gradient Sign Method), IFGSM (Iterative Fast Gradient Sign Method), MIFGSM (Momentum Iterative Fast Gradient Sign Method), Deepfool and Onepixel. The first section includes applying the FGSM (MIFGSM, IFGSM) attacks on the simple CNN model by gradually increasing the perturbation value (epsilon) to evaluate the accuracy variations of the model. It is shown in Table I that lists the epsilon values and CNN model corresponding accuracies (in percentage). For each model accuracy testing, the same training data was used through all 10 epochs.

TABLE I
SUCCESS RATES (%) OF NON-TARGETED ADVERSARIAL ATTACKS
AGAINST SIMPLE CNN MODEL.

Epsilon values	0	0,005	0,015	0,03	0,06	0,10
FGSM	67,80	42,50	13,80	3,06	0,30	0,20
MIFGSM	67,80	41,04	9,22	4,69	0,00	0,00
IFGSM	67,80	40,68	9,09	4,90	0,00	0,00

When applying the adversarial attacks (FGSM, MIFGSM, IFGSM) on the simple CNN model, accuracy gradually decreased from 67,80 % to almost 0,00 % in classifying the images. From the Table I results, we can see that the variations MIFGSM and IFGSM have a greater influence on the accuracy of the model. Perturbation is set from 0,00 to 0,10 among

all adversarial attacks, with pixel value in range [0, 1]. The number of iterations is 10 for I-FGSM and MI-FGSM, and the decay factor μ is 1.0. With moderate values of epsilon, FGSM quickly degraded model accuracy, but it was more detectable due to its higher impact on pixel values. By applying perturbations iteratively, IFGSM achieved a stronger and more reliable misclassification effect, especially at lower epsilon values where FGSM might still struggle. The momentum term allowed MIFGSM to outperform both FGSM and IFGSM, especially in terms of consistent misclassification. The model accuracy under MIFGSM dropped nearly to zero at higher epsilon values, indicating that MIFGSM is an effective adversarial approach for inducing misclassification with smaller visual changes. From the table, we can observe that MI-FGSM remains as a strongest adversary, and together with I-FGSM can attack CNN model with 100 % success rate.

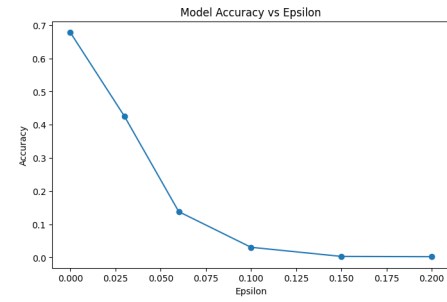


Fig. 3. Trajectory of model accuracy based on different epsilon values.

As we could see from Fig. 4, it is evident that all three attacks affected CNN model, even with a value of ϵ as low as 0.06. We have come to a conclusion that the only aggravating circumstance for attackers when using these attacks (FGSM, IFGSM and MIFGSM) is that with images with a lower resolution, it becomes visible to the naked eye that there have been some changes (perturbations) even with a very small epsilon value (0,06).

The second section includes applying the Deepfool and Onepixel attacks on the simple CNN model. Deepfool and Onepixel attack generate adversarial perturbations which are hardly perceptible (Fig. 5 and 6). DeepFool and Onepixel attacks are computationally efficiently and converge in 10 iterations to a perturbation vector that fools the model classifier. We came to the conclusion that compared to the FGSM attack and its variants, Deepfool and Onepixel are much more efficient and faster attacks whose perturbations are not visible to the naked eye. Deepfool succeeded in producing minimal perturbations that often remained visually imperceptible, showing how slight changes can fool CNNs without noticeable distortions. This feature makes Deepfool an excellent example of a “stealth” attack. OnePixel attack underscores vulnerabilities in CNN models used in imaging, where even a small pixel alteration can lead to incorrect diagnoses or classifications. The findings point to a critical need for adversarial training and defense mechanisms, especially in applications where imperceptible changes, like those in the Deepfool and One-Pixel

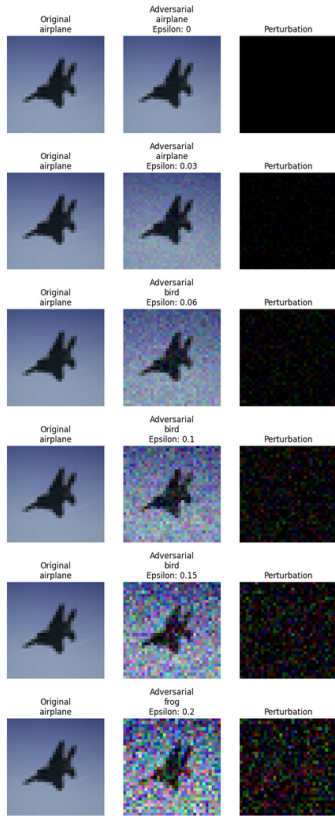


Fig. 4. Effect of epsilon on model accuracy.

Attack, can impact the system's integrity. Future work should focus on evaluating these attacks across different datasets and resolutions to develop more robust, generalizable defenses.

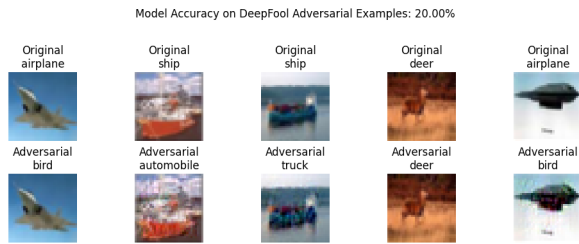


Fig. 5. Effect of DeepFool attack on model accuracy.

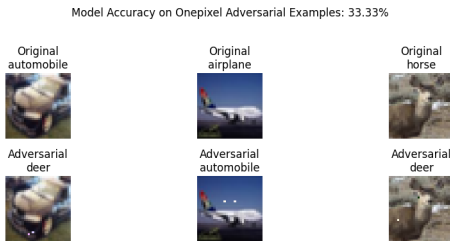


Fig. 6. Effect of OnePixel attack on model accuracy.

V. CONCLUSION

This study highlights the susceptibility of CNN models to adversarial attacks, demonstrating substantial accuracy reductions under different perturbation levels. The findings reveal that while MIFGSM and IFGSM maximize accuracy degradation, methods like Deepfool and OnePixel attack offer imperceptible alterations, making them harder to detect visually. These vulnerabilities emphasize the need for robust defense mechanisms and adversarial training. Future research will extend this analysis to higher-resolution images and multiple datasets to assess attack resilience comprehensively and improve CNN defenses against adversarial manipulation.

REFERENCES

- [1] C. Szegedy et al., "Intriguing properties of neural networks," arXiv.org, 2013. <https://arxiv.org/abs/1312.6199>.
- [2] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and Harnessing Adversarial Examples," arXiv.org, 2014. <https://arxiv.org/abs/1412.6572>.
- [3] K. Ren, T. Zheng, Z. Qin, and X. Liu, "Adversarial Attacks and Defenses in Deep Learning," *Engineering*, vol. 6, no. 3, Jan. 2020, doi: <https://doi.org/10.1016/j.eng.2019.12.012>.
- [4] Z. Liu, W. Peng, J. Zhou, Z. Wu, J. Zhang, and Y. Zhang, "MI-FGSM on Faster R-CNN Object Detector," Dec. 2020, doi: <https://doi.org/10.1145/3447450.3447455>.
- [5] A. Kurakin, I. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," arXiv:1607.02533 [cs, stat], Feb. 2017, Available: <https://arxiv.org/abs/1607.02533>.
- [6] S.-M. Moosavi-Dezfooli, A. Fawzi, and P. Frossard, "DeepFool: A Simple and Accurate Method to Fool Deep Neural Networks," 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2016, doi: <https://doi.org/10.1109/cvpr.2016.282>.
- [7] J. Su, D. V. Vargas, and K. Sakurai, "One Pixel Attack for Fooling Deep Neural Networks," *IEEE Transactions on Evolutionary Computation*, vol. 23, no. 5, pp. 828–841, Oct. 2019, doi: <https://doi.org/10.1109/tevc.2019.2890858>.
- [8] N. Papernot, P. McDaniel, and I. Goodfellow, "Transferability in Machine Learning: from Phenomena to Black-Box Attacks using Adversarial Samples" arXiv:1605.07277 [cs], May 2016, Available: <https://arxiv.org/abs/1605.07277>.
- [9] P. Rathore, A. Basak, S. H. Nistala, and V. Runkana, "Untargeted, Targeted and Universal Adversarial Attacks and Defenses on Time Series," 2020 International Joint Conference on Neural Networks (IJCNN), Jul. 2020, doi: <https://doi.org/10.1109/ijcnn48605.2020.9207272>.
- [10] A. D. Akwaboah, "Convolutional Neural Network for CIFAR-10 Dataset Image Classification", ResearchGate, Nov. 13, 2019. <https://www.researchgate.net>.
- [11] K. O'Shea and R. Nash, "An Introduction to Convolutional Neural Networks," arXiv:1511.08458 [cs], Dec. 2015, Available: <https://arxiv.org/abs/1511.08458>.
- [12] S. Albawi, T. A. Mohammed, and S. Al-Zawi, "Understanding of a Convolutional Neural Network," 2017 International Conference on Engineering and Technology (ICET), pp. 1–6, Aug. 2017, doi: <https://doi.org/10.1109/icengtechnol.2017.8308186>.
- [13] L. Alzubaidi et al., "Review of deep learning: concepts, CNN architectures, challenges, applications, future directions," *Journal of Big Data*, vol. 8, no. 1, Mar. 2021, Available: <https://journalofbigdata.springeropen.com/articles/10.1186/s40537-021-00444-8>.
- [14] Sensitivity of Adversarial Perturbation in Fast Gradient Sign Method — IEEE Conference Publication — IEEE Xplore, ieeexplore.ieee.org/document/9002856.
- [15] A. Kurakin, I. Goodfellow, and S. Bengio, "Adversarial Machine Learning at Scale," arXiv:1611.01236 [cs, stat], Feb. 2017, Available: <https://arxiv.org/abs/1611.01236>.